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1. Multiple-Choice Questions (36)

1. B
2. A
3. C
4. B
5. D
6. A
7. A
8. D
9. A
10. B
11. B
12. D
13. A
14. C
15. A
16. B
17. C
18. B
19. A
20. C
21. D
22. A
23. B
24. C
25. B
26. A
27. C
28. D

29. B
30. A
31. C
32. D
33. A
34. B
35. A
36. C

2. True or False (61)

1. True
2. False
3. True
4. True
5. False
6. True
7. False
8. True
9. True
10. False
11. True
12. True
13. True
14. True
15. False
16. True
17. True
18. True
19. False
20. False
21. True
22. True
23. True
24. True
25. True
26. False

27. False
28. True
29. True
30. True
31. True
32. False
33. True
34. True
35. True
36. False
37. True
38. True
39. False
40. True
41. True
42. True
43. False
44. True
45. False
46. True
47. True
48. True
49. True
50. True
51. False
52. True
53. True
54. True
55. True
56. False
57. False
58. False
59. True
60. False
61. True

3. Fill in the Blanks (51)

1. Data
2. Requirements
3. Conceptual
4. Logical
5. Physical
6. Logical
7. Physically
8. Needs
9. Entity
10. Attribute
11. Entity
12. Attributes
13. Attribute
14. Entities
15. Structure
16. Entity
17. Attribute
18. Relationship
19. Cardinality
20. Constraints
21. Entity
22. Attribute
23. Entity
24. Attributes
25. Attribute
26. Entities
27. Visual
28. Cardinality
29. Entities
30. Attributes
31. Attributes
32. Relationship
33. Understand
34. Tables
35. Student

36. Course
37. Relationship
38. Attributes
39. Attributes
40. Tables
41. Entities
42. Enrolls
43. Places
44. Works
45. Business
46. Cardinality
47. One-to-One (1:1)
48. One-to-Many (1:M)
49. Many-to-Many (M:N)
50. Mandatory
51. 1:M

4. Short-Answer Questions (60)

1. The process of deciding what data to store and how to organize it.
2. Requirements analysis.
3. Conceptual design.
4. Logical design.
5. Physical design.
6. Requirements Analysis → Conceptual Design → Logical Design → Physical Design.
7. To identify users' needs and data requirements.
8. Logical design.
9. Requirements analysis, conceptual design, logical design, physical design.
10. Understanding users' needs and system requirements.
11. Creating a high-level model of data and relationships.
12. Defining the logical structure of the database.
13. Deciding how data is physically stored.
14. Conceptual design.
15. Logical design.
16. Physical design.
17. It organizes data, reduces redundancy, and improves data integrity.
18. Describing data structure and relationships.

19. An important object about which information is stored.
20. A property that describes an entity.
21. How many instances of entities can participate in a relationship.
22. Rules that control valid data.
23. They represent connections between entities and business rules.
24. They ensure data is valid and consistent.
25. Student, Course, Employee, Customer, Product.
26. Student_ID, Name, Email, Phone.
27. Student.
28. Student_ID, Name, Email, Phone.
29. An entity is an object; an attribute is a property of that object.
30. Student, Course, Employee, Customer, Product.
31. Student_ID, Name, Email, Phone.
32. Student.
33. Student_ID, Name, Email, Phone.
34. An entity is an object; an attribute describes it.
35. A visual representation of an ER model.
36. Entities, attributes, relationships, and cardinality.
37. STUDENT and COURSE.
38. Student_ID and Name.
39. Course_ID and Course_Name.
40. Enrolls.
41. They make database requirements easier to understand and discuss.
42. Converting an ER model into relational tables.
43. Student(Student_ID, Name, Email).
44. Course(Course_ID, Course_Name).
45. Enrollment(Student_ID, Course_ID).
46. To represent the relationship between Student and Course.
47. Attributes or fields.
48. Student and Course represent entities; Enrollment represents their relationship.
49. 1:1, 1:M, M:N.
50. One instance relates to exactly one instance of another entity.
51. One instance relates to many instances of another entity.
52. Many instances of one entity relate to many instances of another.
53. Participation that is required.
54. Participation that is not required.
55. A connection between two or more entities.
56. Student enrolls in Course.

57. Customer places Order.
58. Employee works in Department.
59. They represent real-world connections and business rules.
60. Two or more entities can participate in a relationship.

5. Explain Why Questions (40)

1. To understand users' needs and avoid missing requirements.
2. To create a high-level view of entities and relationships.
3. To define the database structure before implementation.
4. To determine efficient physical storage and access.
5. To organize the process and reduce errors.
6. To organize data and relationships clearly.
7. To identify the important objects the database must store.
8. To describe the properties of entities.
9. To define how many instances can participate in a relationship.
10. To ensure data validity and consistency.
11. To identify the objects about which information is stored.
12. To describe entity properties.
13. Student is an object; Name describes the student.
14. Because it identifies or describes a student.
15. To model the database correctly.
16. To identify the objects the system must store.
17. To describe information about entities.
18. Student is an object; Name is its property.
19. Because it identifies or describes a student.
20. To distinguish objects from their properties.
21. Because it presents the ER model graphically.
22. To identify the objects being represented.
23. To show the properties of entities.
24. To clarify the number of related instances.
25. They make requirements easier to understand and discuss.
26. To implement the model in a relational database.
27. Because the Student entity stores student information.
28. Because the Course entity stores course information.
29. To represent the relationship between Student and Course.
30. It converts a conceptual model into a usable database structure.

31. To represent connections between entities.
32. Because students actually enroll in courses.
33. It represents how customers place orders.
34. It identifies which department employees belong to.
35. To implement the correct database structure and business rules.
36. To determine the number of related instances.
37. To specify whether participation is required or optional.
38. Because one department can contain many employees.
39. Because one student can take many courses and one course can have many students.
40. To implement the correct relationships and enforce business rules.

6. Ordering Questions (3)

1. Requirements Analysis → Conceptual Design → Logical Design → Physical Design.
2. Conceptual Model → Logical Design.
3. Logical Design.

7. Scenario-Based Questions (49)

1. Requirements analysis.
2. Users' needs, data requirements, and business rules.
3. A conceptual ER model.
4. The logical database structure is defined.
5. Physical storage and implementation are determined.
6. Students and Courses.
7. Student_ID and Name.
8. Student enrolls in Course.
9. How many students can take courses and vice versa.
10. Student_ID must be unique.
11. Student.
12. Student_ID, Name, Email, Phone.
13. Employee.
14. Employee_ID, Name, Salary.
15. Product.
16. Product_ID, Name, Price.

17. Student.
18. Student_ID, Name, Email, Phone.
19. Employee.
20. Employee_ID, Name, Salary.
21. Product.
22. Product_ID, Name, Price.
23. STUDENT and COURSE.
24. Enrolls.
25. Student_ID and Name.
26. Course_ID and Course_Name.
27. To clarify entities, attributes, relationships, and cardinality before implementation.
28. Cardinality.
29. Because it represents the model graphically.
30. To show the important objects.
31. To describe entity properties.
32. To show the number of related instances.
33. They make database requirements easier to understand.
34. Student, Course, Enrollment.
35. Student_ID, Name, Email.
36. Course_ID, Course_Name.
37. Student_ID, Course_ID.
38. It represents the Student–Course relationship.
39. Enrollment.
40. Student enrolls in Course.
41. Customer places Order.
42. Employee works in Department.
43. Doctor treats Patient.
44. Member borrows Book.
45. 1:M.
46. M:N.
47. 1:1.
48. Mandatory participation.
49. Optional participation.

8. Compare and Contrast (30)

1. Requirements Analysis: Identifies needs. Conceptual Design: Creates a high-level data model.
2. Conceptual Design: High-level model. Logical Design: Defines tables, keys, and relationships.
3. Logical Design: Defines database structure. Physical Design: Defines storage and implementation.
4. Conceptual Design: High-level representation. Physical Design: Storage and performance details.
5. Entity: An object. Attribute: A property of an object.
6. Relationship: Connection between entities. Cardinality: Number of related instances.
7. Attribute: Describes an entity. Constraint: Restricts valid data.
8. Entity: An object. Attribute: A property.
9. Student: Entity. Name: Attribute.
10. Customer: Entity. Email: Attribute.
11. Product: Entity. Price: Attribute.
12. Entity: An object. Attribute: A property.
13. Student: Entity. Name: Attribute.
14. Customer: Entity. Email: Attribute.
15. Product: Entity. Price: Attribute.
16. ER Model: Conceptual representation of entities and relationships. ER Diagram: Visual representation of that model.
17. Entity: Object. Attribute: Property.
18. Relationship: Connection. Cardinality: Number of related instances.
19. ER Diagram: Design representation. Implemented Database: Actual database storing data.
20. ER Model: Conceptual data representation. Relational Database: Data organized into tables.
21. Student Table: Stores student details. Enrollment Table: Stores student-course associations.
22. Entity Table: Stores entity data. Relationship Table: Stores associations, especially M:N relationships.
23. Student_ID in Student: Primary key. Student_ID in Enrollment: Foreign key referencing Student.

24. Both represent relationships between real-world entities; one concerns students and courses, the other customers and orders.
25. Entity: An object. Relationship: A connection between entities.
26. Relationship: Connects entities. Attribute: Describes an entity.
27. 1:1: One instance relates to one. 1:M: One instance relates to many.
28. 1:M: One-to-many. M:N: Many-to-many.
29. Cardinality: Number of related instances. Participation: Whether involvement is mandatory or optional.
30. Mandatory: Participation is required. Optional: Participation is not required.

9. Higher-Order Thinking Questions (40)

1. Requirements may be missed, causing an incorrect database design.
2. Errors, rework, and an inefficient database structure may result.
3. It establishes the high-level structure before detailed design.
4. Physical storage decisions require a defined logical structure.
5. Requirements analysis and the relevant design stages may need revision.
6. Redundancy, inconsistency, missing relationships, and poor organization.
7. Important information cannot be stored or represented correctly.
8. The database may allow invalid relationships or restrict valid ones.
9. To prevent duplicate or invalid student identifiers.
10. Entities contain attributes and connect through relationships.
11. Student.
12. The Phone attribute is removed from the Student model or table.
13. Yes.
14. Yes.
15. Student_ID can uniquely identify a student, unlike Name.
16. Student.
17. The Phone attribute is removed.
18. Yes.
19. Yes.
20. Student_ID is more reliable for unique identification.
21. Connections between entities become unclear or missing.
22. The student identifier and name information would be missing.
23. It is visual, easier to understand, and does not require reading code.
24. Add the LIBRARY entity, its attributes, and relevant relationships.
25. The relationship's cardinality notation changes.

26. They identify the student and course participating in each enrollment.
27. The information connecting students to courses may be lost.
28. The Student table.
29. The Course table.
30. By converting entities into tables, attributes into columns, and relationships into keys or relationship tables.
31. Related data cannot be represented or retrieved correctly.
32. Which students take which courses.
33. The employee's department association must be updated.
34. Yes.
35. They specify valid connections and enforce business rules.
36. One-to-many (1:M), with mandatory employee participation.
37. Student participation is optional.
38. Mandatory participation.
39. The database may fail to represent multiple valid associations correctly.
40. Yes. Cardinality defines how many instances are related; participation defines whether the relationship is required.